

LeSuccess Academy — Chatbot Architecture Guide

Clarification on Normal Chatbot & Re-enabling the API-Based Chatbot

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Executive Summary

The LeSuccess Academy website currently features a **dual-chatbot architecture**:

- Existing API-Based Chatbot (AI Powered)**: Uses Spring AI, Google Gemini (`gemini-3.8-flash`), embeddings, and backend chat controllers. It has been **safely preserved in full** and only temporarily hidden from the frontend UI.
- New Normal Chatbot (Rule-Based / FAQ)**: A 100% client-side, zero-dependency assistant that answers visitor questions instantly with predefined knowledge and keyword matching. It requires **no API keys, no external AI services, and no backend connectivity**.

1. Re-Enabling the Existing API-Based Chatbot

Because the entire API-based chatbot implementation was preserved intact across the frontend and backend, restoring it requires updating only **one single file** (`frontend/src/App.jsx`).

Step 1: Swap the Import in `frontend/src/App.jsx`

Open `frontend/src/App.jsx`. Near **Line 17**, switch the active chatbot import:

```
// Comment out or remove the BasicChatbot import:
// import BasicChatbot from "../components/chatbot/BasicChatbot.jsx";

// Uncomment the original ChatWidget import:
import ChatWidget from "../components/chatbot/ChatWidget.jsx";
```

Step 2: Swap the Component Rendering in `frontend/src/App.jsx`

Scroll down to the chatbot section near **Line 595**:

```
{/* Render the original API-based ChatWidget: */}
{!isDashboard && !isLoginPage && (
  <ChatWidget onOpenEnquiry={() => setIsEnquiryOpen(true)} />
)}
```

Step 3: Verify Environment Variables & Backend Service

The underlying configuration is already fully set up in your project:

- Frontend** (`frontend/.env.development`):
 - `VITE_CHATBOT_ENABLED=true` (Controls whether the widget displays; already configured as true)
- Backend** (`.env` for local development, or **Render Environment Variables** for production):
 - `CHATBOT_ENABLED=true`
 - `GEMINI_API_KEY=<your-google-gemini-api-key>`
 - `GEMINI_MODEL=gemini-3.8-flash`
- Backend Status**: Ensure the Spring Boot backend server is running (`mvn spring-boot:run` or `./scripts/run-dev.ps1`) so it can receive `POST /api/chat` requests.

Verification of Preserved Components

The following files remain completely untouched and ready in your codebase:

- `frontend/src/components/chatbot/ChatWidget.jsx` (Launcher & lazy window wrapper)
- `frontend/src/components/chatbot/ChatWindow.jsx` (Conversation interface & markdown renderer)
- `frontend/src/components/chatbot/ChatLauncher.jsx` (Floating circular launcher with logo GIF)
- `frontend/src/hooks/useChat.js` (Session state & message lifecycle manager)
- `frontend/src/services/chatApi.js` (HTTP client communication with backend `/api/chat`)
- `backend/.../ChatController.java` & `ChatService.java` (Spring Boot API endpoints & Gemini integration)

2. New Normal Chatbot — Technical Clarification

The newly created **Normal Chatbot** (`BasicChatbot.jsx`) was custom-built to operate completely independently of any AI provider or external server.

Direct Answers to Key Questions

Evaluation Question	Answer	Technical Details
Does it require an API key?	NO	Contains no references to API keys and requires none to function.
Does it make external AI/API calls?	NO	Never contacts Google Gemini, OpenAI, Claude, or any external service.
Does it require a backend server?	NO	Operates 100% in the user's browser. Works even if the backend is stopped.
Does it require .env API keys?	NO	Reads no environment variables for AI keys, models, or tokens.
Can it function offline?	YES	Once loaded in the browser, all queries are processed locally in JavaScript.

How the Normal Chatbot Generates Responses

The Normal Chatbot uses an **Intelligent Client-Side Intent Matching Engine**:

- Text Normalization**: When a visitor enters a message, punctuation and whitespace are cleaned and converted to lowercase tokens.
- Intent & Keyword Matching**: The engine checks the input against a structured academy knowledge base covering:
 - Greetings**: "hi", "hello", "hey", "good morning"
 - Courses Overview**: "courses", "what courses do you offer", "curriculum"
 - Full Stack Java**: Core/Advanced Java, Spring Boot, Microservices, Hibernate
 - Python Full Stack**: Python programming, Django, FastAPI, React frontend
 - MERN Full Stack**: MongoDB, Express.js, React.js, Node.js
 - Data Analytics & Science**: Power BI, Tableau, Advanced SQL, Python, Pandas
 - AWS Cloud & DevOps**: EC2, Docker, Kubernetes, CI/CD pipelines
 - Placements & Internships**: 100% placement support, mock interviews, 50+ hiring partners
 - Contact & Location**: Phone, WhatsApp (+91 99622 28445), Email, Chennai location
 - Timings & Schedules**: Weekday & weekend batches, online & classroom options
 - Fees & Scholarships**: Flexible installments, 0% EMI options, counseling guidance
 - Webinars & Workshops**: Live interactive bootcamps and certification events
- Interactive Action Links**: Answers include direct on-site navigation pills (such as links to `/courses`, `/contact`, `/programs`, or the enquiry form modal).
- Deterministic Fallback**: If an input does not match any known topic, it displays a friendly fallback:

"I'm currently a basic assistant, so I can help with LeSuccess courses, programs, contact information, registration and other common questions. Please try asking about one of these topics, or contact our team directly!"

3. Architecture Comparison Summary

Feature	Existing API-Based Chatbot	New Normal Chatbot
Component Name	<code>ChatWidget / ChatWindow</code>	<code>BasicChatbot</code>
Response Engine	Google Gemini Generative AI Model	Client-Side Rule & Intent Engine
API Key Required	Yes (<code>GEMINI_API_KEY</code>)	No API Key Required
Network Requests	<code>POST /api/chat</code> to Spring Boot	Zero Network Requests
Hosting Cost	Dependent on AI Token Usage	100% Free & Unlimited
Response Latency	Network & Model Dependent (~1-3s)	Instantaneous (~350ms natural simulation)
Reliability	Depends on AI API availability & quota	Deterministic & 100% Available
Current Status	Preserved in codebase (Hidden in <code>App.jsx</code>)	Active on Website

Prepared for LeSuccess Academy Management & Development Team.